One can write the coefficients of the variables by considering all the missing terms in the standard form of the polynomial. For example : $x^3 - 3x^2 + 0x - 8$ can be written as $(1, -3, 0, -8)$. This form of the polynomial is called **Coefficient form.**

Polynomial $(4, 0, -5, 0, 1)$ can be written by using variable y as

$4y^4 + 0y^3 - 5y^2 + 0y + 1$. This form is called **Index form** of the polynomial.

Ex. $p(m) = 3m^5 - 7m + 5m^3 + 2$

Write the polynomial in standard form	$3m^5 + 5m^3 - 7m + 2$
Write it in the index form by considering all the missing terms with coefficient zero.	$3m^5 + 0m^4 + 5m^3 + 0m^2 - 7m + 2$
Write it in a coefficient form	$(3, 0, 5, 0, -7, 2)$
Degree of the polynomial	5

Ex (1) Write the polynomial $x^3 + 3x - 5$ in coefficient form.

Solution : $x^3 + 3x - 5 = x^3 + 0x^2 + 3x - 5$

$\therefore$ given polynomial in coefficient form is $(1, 0, 3, -5)$

Ex (2) $(2, -1, 0, 5, 6)$ is the coefficient form of the polynomial. Represent it in index form.

Solution : Coefficient form of the polynomial is $(2, -1, 0, 5, 6)$

$\therefore$ index form of the polynomial is $2x^4 - x^3 + 0x^2 + 5x + 6$ i.e. $2x^4 - x^3 + 5x + 6$

Practice set 3.1

1. State whether the given algebraic expressions are polynomials ? Justify.

   (i) $y + \frac{1}{y}$ (ii) $2 - 5\sqrt{x}$ (iii) $x^2 + 7x + 9$

   (iv) $2m^{-2} + 7m - 5$ (v) 10

2. Write the coefficient of m^3 in each of the given polynomial.

   (i) m^3 (ii) $\frac{-3}{2} + m - \sqrt{3}\,m^3$ (iii) $\frac{-2}{3}m^3 - 5m^2 + 7m - 1$

3. Write the polynomial in x using the given information.

   (i) Monomial with degree 7 (ii) Binomial with degree 35

   (iii) Trinomial with degree 8

4. Write the degree of the given polynomials.

(i) $\sqrt{5}$ (ii) x° (iii) x^2 (iv) $\sqrt{2}m^{10} - 7$ (v) $2p - \sqrt{7}$

(vi) $7y - y^3 + y^5$ (vii) $xyz + xy - z$ (viii) $m^3n^7 - 3m^5n + mn$

5. Classify the following polynomials as linear, quadratic and cubic polynomial.

(i) $2x^2 + 3x + 1$ (ii) $5p$ (iii) $\sqrt{2}\ y - \frac{1}{2}$

(iv) $m^3 + 7m^2 + \frac{5}{2}m - \sqrt{7}$ (v) a^2 (vi) $3r^3$

6. Write the following polynomials in standard form.

(i) $m^3 + 3 + 5m$ (ii) $-7y + y^5 + 3y^3 - \frac{1}{2} + 2y^4 - y^2$

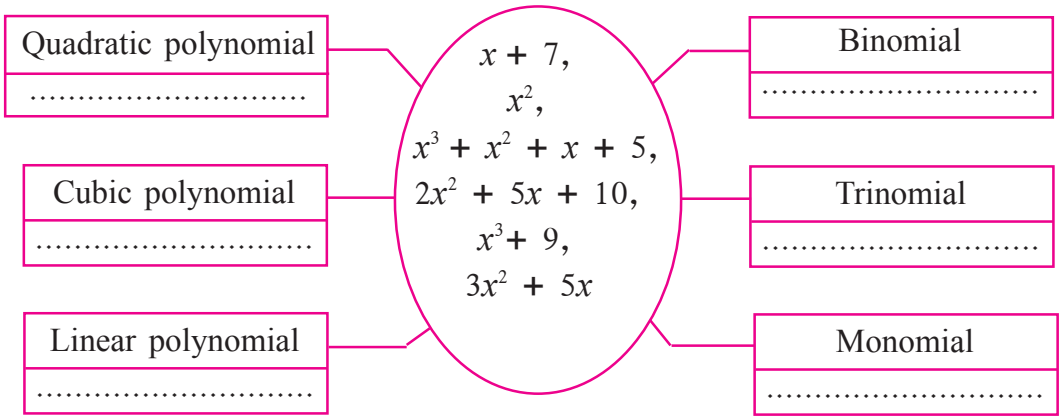
7. Write the following polynomials in coefficient form.

(i) $x^3 - 2$ (ii) $5y$ (iii) $2m^4 - 3m^2 + 7$ (iv) $-\frac{2}{3}$

8. Write the polynomials in standard form.

(i) (1, 2, 3) (ii) (5, 0, 0, 0, – 1) (iii) (– 2, 2, – 2, 2)

9. Write the appropriate polynomials in the boxes.





Let's recall.

(1) Coefficients are added or subtracted while adding or subtracting like algebraic terms, e.g. $5m^3 - 7m^3 = (5 - 7)m^3 = -2m^3$

(2) While multiplying or dividing two algebraic terms, we multiply or divide their coefficients. We also use laws of indices.

$-4y^3 \times 2y^2z = -8y^5z$; $12a^2b \div 3ab^2 = \frac{4a}{b}$